Task 6 : Create a Strong Password and Evaluate Its Strength

Prepared By: Arti Kumari

Tools Used: https://passwordmeter.com/

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Objective

- To understand what makes a password strong and test it against password strength tools

Requirements for a good password

- Minimum 8 characters in length
- Contains 3/4 of the following items:
  - Uppercase Letters
  - Lowercase Letters
  - Numbers
  - Symbols

Password Used : It"s0SiMpLe

Common Password attacks

Brute-Force Attack

- **Method:** The attacker systematically tries **every single possible combination** of characters (letters, numbers, symbols) until the correct password hash is found.
- **Defense:** This attack is defeated by **password length** and **entropy**. A longer, more complex password exponentially increases the time required, making the attack computationally infeasible.

Dictionary Attack

- **Method:** The attackeruses a predefined list (a "dictionary") of the most common passwords, words, phrases, and substitution patterns (like "p@ssw0rd") and tests them against the stolen password hashes.
- **Defense:** Use **unique passphrases** composed of unrelated words or use characters not found in any common language.

Hybrid Attack

- **Method:** A combination of Brute-Force and Dictionary attacks. The attacker takes common words from a dictionary (e.g., `Summer2025`) and then systematically adds numbers and symbols to the beginning or end (e.g., `!Summer2025`, `Summer2025!!`, `Smmr2025!`).
- **Defense:** Avoid using names, dates, or common phrases in your passwords, even if you substitute some characters.

Observation

Test Your Password		Minimum Requirements
Password:	It"s0SiMpLe	• Minimum 8 characters in length • Contains 3/4 of the following items: - Uppercase Letters - Lowercase Letters - Numbers - Symbols
Hide:		
Score:	94%	
Complexity:	Very Strong	

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	+(n*4)	11	+ 44
Uppercase Letters	Cond/ Incr	+((len-n)*2)	4	+ 14
Lowercase Letters	Cond/ Incr	+((len-n)*2)	5	+ 12
Numbers	Cond	+(n*4)	1	+ 4
Symbols	Flat	+(n*6)	1	+ 6
Middle Numbers or Symbols	Flat	+(n*2)	2	+ 4
Requirements	Flat	+(n*2)	5	+ 10
Deductions				
Letters Only	Flat	-n	0	0
Numbers Only	Flat	-n	0	0
Repeat Characters (Case Insensitive)	Comp	-	0	0
Consecutive Uppercase Letters	Flat	-(n*2)		

Conclusion

This task underscored that the objective of strong password creation is to make the computational effort required for guessing or cracking the password **exceed the economic and time limits** of any attacker.